

the origins of data

what are data and where do they come from?

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a short introduction

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production

media studies, software studies, feminist epistemology,
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a short introduction to Critical Data Studies

- computational turn → transformation within society with the introduction of electronic computation
- datafication of everything
- what are data?

emergence of CDS

- emerged as a field in the 2010s
- highly interdisciplinary
- technology as never as neutral as it appears
- data as never raw
- key themes → power and politics

practising Critical Data Studies

- how data are produced, circulated, and utilized
- “post positivist” framing
- against the end of theory
- aim to study data assemblages
 - the technological, political, social and economic apparatuses and elements that constitutes and frames the generation, circulation and deployment of data
- asking reflexive and challenging questions about data
- seeks to challenge and transform data power

data ethnographies and auto-ethnographies; interviews, focus groups and workshops; data archaeology and traces; historical analysis and genealogies; participatory and action research; data walks and derives; discursive analysis; research-creation and creative practice; critical data science

System/process
performs a task

Reception/operation
(user/usage)

Interface

Code/algorithms
(software)

Data(base)

Code platform
(operating system)

Material platform
(infrastructure – hardware)

Context
frames the system/task

Systems of thought

Modes of thinking, philosophies, theories, models, ideologies, rationalities, etc.

Forms of knowledge

Research texts, manuals, magazines, websites, experience, word of mouth, etc.

Finance

Business models, investment, venture capital, grants, philanthropy, profit, etc.

Political economies

Policy, tax regimes, incentive instruments, public and political opinion, etc.

Governmentalities and legalities

Data standards, file formats, protocols, regulations, laws, licensing, IPR, etc.

Materialities and infrastructures

Paper/pens, computers, sensors, databases, networks, servers, buildings, etc.

Practices

Techniques, ways of doing, learned behaviours, scientific conventions, etc.

Organisations and institutions

Corporations, government agencies, NGOs, universities, civil groups, etc.

Subjectivities and communities

Of data producers, curators, managers, analysts, politicians, users, citizens, etc.

Places

Labs, offices, field sites, data centres, business parks and their agglomerations

Marketplace

For data, their derivatives, analysts, analytic software, interpretations, etc.

processes of classification

- classification as simplification / as making sense of the world
- classification as not naturally occurring
- considering ubiquity, materiality, indeterminacy of past, practical politics
- what is counted often becomes the basis for policymaking and resource allocation
- paradox of exposure
- matrix of domination
- questions about counting must be accompanied by questions about consent, as well as of personal safety, cultural dignity, and historical context
 - sometimes the most ethical thing to do is to help people be obscure, hidden, and invisible
- historically, counting and classification have been used to dominate, discipline, and exclude

the big data paradigm

big data: data that are **generated and processed** in real-time and are exhaustive to a system; that is, they are **continuously produced** for an entire population rather than a sample (*characterized by volume, velocity, variety, exhaustivity, granularity, indexicality, relationality, extensionality, scalability*)

paradigm: ???

what is a paradigm?

*“[...]a paradigm constitutes an accepted way of
interrogating the world and synthesising knowledge
common to a substantial proportion of researchers in a
common discipline[...]”*

(Kitchin, 2021: 113, paraphrasing Kuhn, 1962)

what is a paradigm?

looking at the natural sciences:

- the scientific method
- hypothesis testing
- verification / falsification



- neither static nor permanent
- change over time
- BUT are temporarily hegemonic

big data = a new empiricism?

Claim:

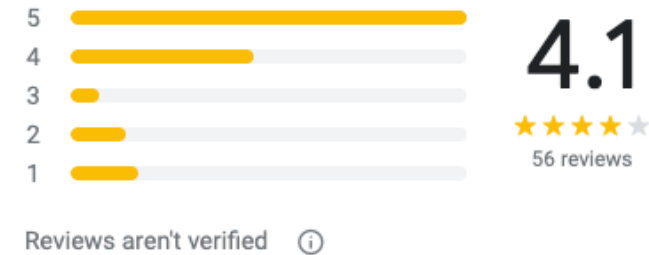
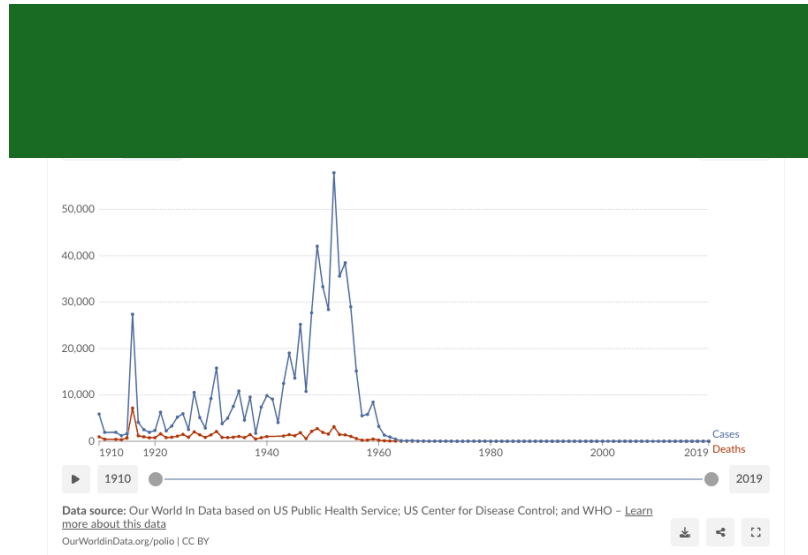
data 'can reveal their inherent truth' and
'speak for themselves' (Kitchin, 2021). Correlation
beats causation. Induction trumps deduction. No need
to ask questions, theory is dead.

the fallacies of big data empiricism

1. big data are **never “complete”** and always sampled
2. big data do not materialise in a **vacuum**
3. big data **are not free** from human bias
4. big data are **meaningless without context** and domain knowledge

the fallacies of big data empiricism

1. scraping all user data on Instagram?
2. leaving a review?
3. what are algorithmic healthcare and insurance decisions based on?
4. why does this chart need context?



w2 literature

Critical data studies: An introduction

(Andrew Illiadis¹ and Federica Russo², 2016)

¹ **Andrew Illiadis** is currently an associate professor in media and communication at Temple University (US). Illiadis' work focuses on the social implications of data science with specific interests in online search and knowledge sharing, wearable technologies and other embodied computing devices, and automated decision-making in policing, intelligence, and national defence.

² **Federica Russo** is a full professor in philosophy and the ethics of techno-science at Utrecht University (NL). Russo's research focuses on epistemological, methodological, and normative aspects that arise in the health and social sciences, with special attention to policy contexts and to the highly technologized character of these fields.

which sentence was the most familiar to you while reading?
which sentence seemed most unfamiliar to you?

how does CDS imagine data? does this image compliment or clash with
your perspective / your discipline's perspective?

how do you perceive the relationship between data and theory?
how do you work with theory?

is having more data always beneficial?
could you work with 'small data'?

could you include the orientations and principles of CDS in your work?
if so, where? if not, why would it not be possible?

further reading recommendations

Boyd, D., & Crawford, K. (2012). [CRITICAL QUESTIONS FOR BIG DATA: Provocations for a cultural, technological, and scholarly phenomenon](#). *Information, Communication & Society*, 15(5), 662–679.

Couldry, N., & Mejias, U. A. (2019). [Data Colonialism: Rethinking Big Data's Relation to the Contemporary Subject](#). *Television & New Media*, 20(4), 336–349.

D'Ignazio, C., & Klein, L. F. (2020). [Data Feminism](#). MIT Press.

Kennedy, H., Hill, R. L., Aiello, G., & Allen, W. (2016). [The work that visualisation conventions do](#). *Information, Communication & Society*, 19(6), 715–735.

Kitchin, R. (2025). [Critical Data Studies: An A to Z Guide to Concepts and Methods](#). Polity Press.

Loukissas, Y. A., & Bowker, G. C. (2019). [All data are local: Thinking critically in a data-driven society](#). The MIT Press.